



Tutorial: Sonny Import Module

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Credit

Altimetric data (DEM/ Mesh): Digital terrain models based on Sonny's LiDAR Digital Terrain Models of Europe, available on sonny.4lima.de . Licensee under [Creative Commons Attribution 4.0 \(CC BY 4.0\)](https://creativecommons.org/licenses/by/4.0/).

Presentation

The main difference between a 20-meter Digital Elevation Model (DEM or DTM) and a 1 arc-second (1") model lies in their spatial resolution, i.e., the size of each pixel or grid cell. A 20m DTM has a fixed 20 x 20-meter grid, while a 1" DTM (1 arc-second) corresponds to an angular grid of approximately 30 meters on the ground (at the equator).

Features of the 20m DTM

Grid size: 20 meters by 20 meters.

Unit: Metric (local or national projected system).

Precision: Coarser, ideal for regional or national scale analyses (file size reduction).

Origin: Often derived from the resampling (generalization) of more precise national topographic data (e.g., a 1m DEM).

Features of the 1" (1 arc-second) DTM

Grid size: Approximately 30 meters (1/3600th of a degree, roughly 30m x 30m).

Unit: Geographic coordinates (latitude/longitude).

Precision: Comparable in raw size to 30m, but dependent on a global angular grid (such as USGS SRTM or Copernicus DEM data).

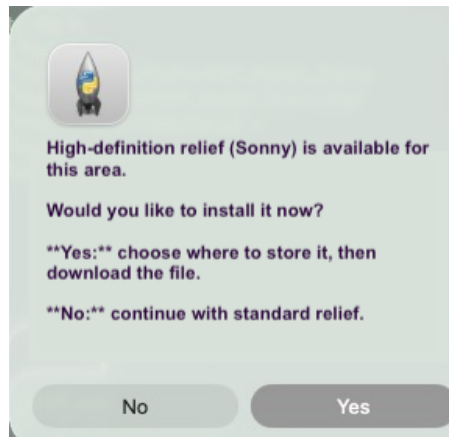
Origin: Designed for global or international coverage of the Earth's surface.

Workflow Étape par Étape

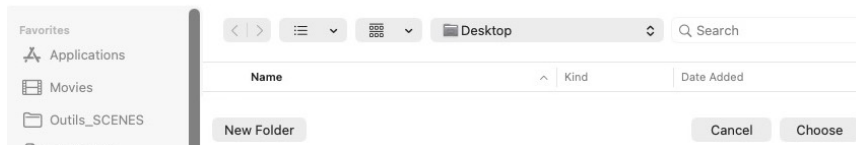
In ORTHO4XP_V3, navigate to Advanced / Altimetry / Click on the "Sonny Relief" button.



Click on "Yes" or "No" according to your choice and follow the instructions.



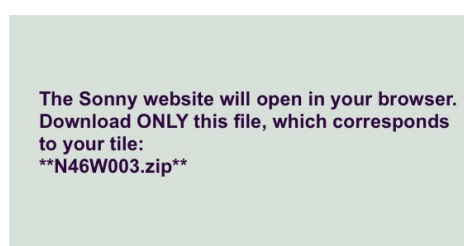
Select your final storage space and click "Choose".



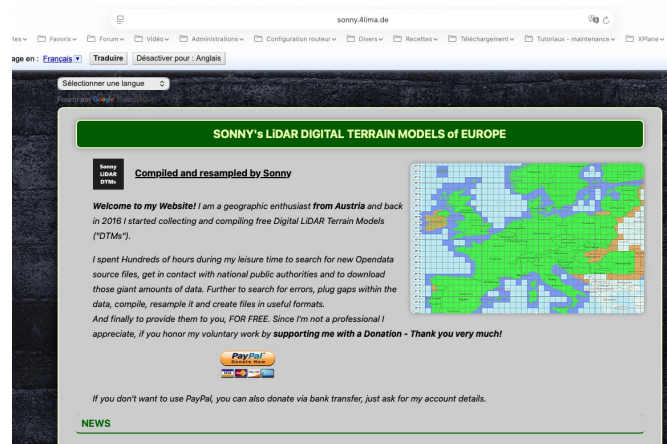
The host structure for your files is created.



In this message, note the number of the file to download, for example: N46W003.zip



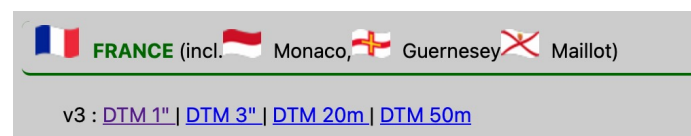
The Sonny site will open in your browser.



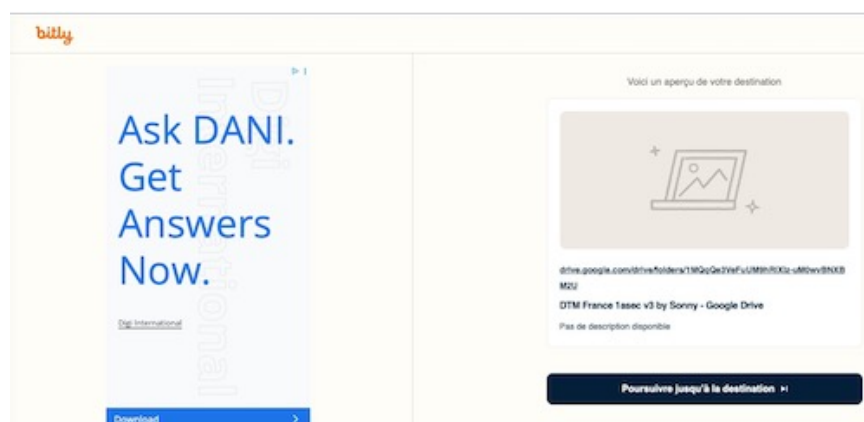
Scroll down on the Sonny page to the "Downloads" section.



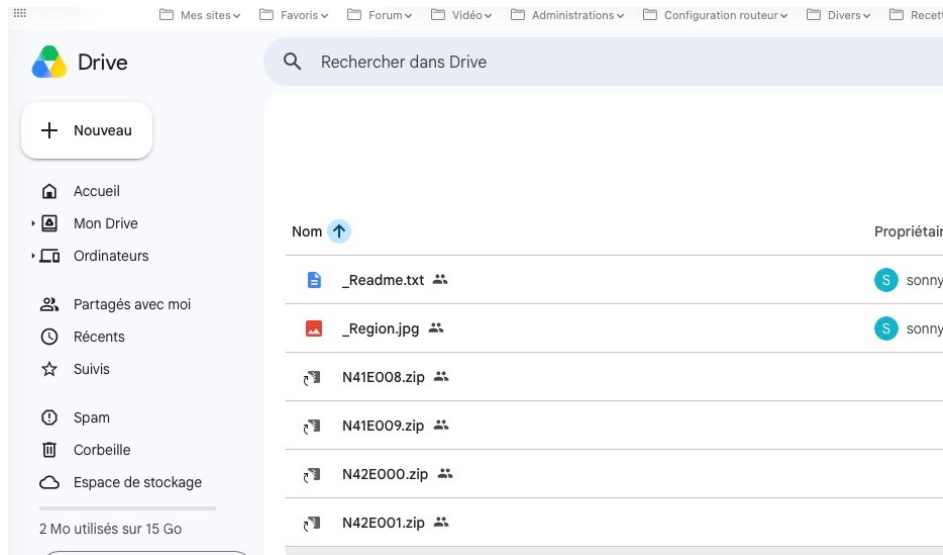
Select the country and click on "DTM 1" under the Country Name (e.g., France).



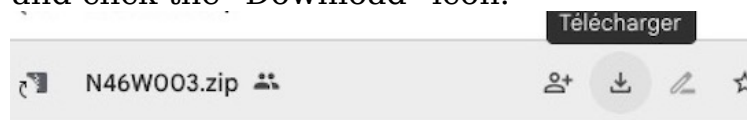
On the following page, click "continue" until you reach the destination.



Search in the list and download the numbered file, for example: N46W003.zip corresponding to the tile currently being created in Ortho.



Select the file row and click the "Download" icon.



Close the browser. Your N46W003.zip file will be found in your system's default Downloads folder.

Click the "Install downloaded ZIP" button.



Close the browser. Your N46W003.zip file will be found in your system's default Downloads folder.

Unzip the N46W003.zip file. It will be placed in the France folder in our example. It is now named "N46W003.hgt".



Return to ORTHO4XP_Vxxx and open the "Tools" window. Click on the "Load tile cfg" or "Reload app cfg" button.



The path to the "N46W003.hgt" file will be displayed in the "custom_dem" field.

